An Exploratory Sequential Deletion Audit of Small Language Models

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An unsuccessful extraction prompt is weak evidence that a language model has forgotten an association. We study this distinction in a small, reproducible adaptation experiment with two sequential deletion requests. Fictional subjects receive six-digit codes; untrained control subjects use exactly the same code inventories as trained subjects, preventing marginal answer familiarity from explaining the inclusion comparison. Development runs first test whether the model learns the actual subject–code associations. Three tested LoRA recipes fail that gate, while a last-four-layer partial-dense recipe passes and is frozen before two corpus/order-seed evaluations. After deleting two groups through cheap retained-only continued adaptation, Qwen3-0.6B produces none of the 16 deleted codes under the exact greedy extraction prompt in either corpus, yet ranks the correct code first for 11/16 deleted subjects. Current-checkpoint association scores distinguish deleted from matched untrained associations with descriptive AUROC 0.8164 and 0.8477. A fixed-recipe replay on Qwen3.5-0.8B gives 1/16 and 0/16 final exact extractions, while 8/16 and 7/16 deleted associations remain top-ranked and current-checkpoint AUROC is 0.7305 and 0.6875. This exploratory replay reuses the same corpora and does not isolate a model-age effect. Exact retained association utility is preserved; paraphrase generation is not matched. Fresh retained-data references and separately labeled prior-snapshot attacks further expose the importance of attacker access. These are controlled pilot observations about an inexpensive baseline, not evidence of LoRA unlearning efficacy, a new unlearning algorithm, or a privacy guarantee.

Additional Key Words and Phrases: machine unlearning, sequential deletion, language models, privacy auditing, preliminary study

1 Introduction and Background

Deleting adaptation records from a training pipeline and preventing their reproduction by a deployed model are related but different goals. A model can cease producing a string under one prompt while retaining a preference for that string over plausible alternatives. Conversely, an adversary retaining an old model may extract information unavailable from the current model alone. An empirical audit must state its observation, candidate knowledge, checkpoint access and utility comparison before interpreting either outcome.

These distinctions are established concerns. TOFU uses fictional author records and multiple measures to study targeted removal [8]. MUSE explicitly separates memorization, privacy, utility, scale and repeated requests [12]. OpenUnlearning supplies a common implementation and metric ecosystem [2]; constructing another general benchmark platform is therefore unnecessary. Du et al. demonstrate textual membership and reconstruction risks after unlearning [3]. Wu et al. show why access to both earlier and later logits changes extraction even after retained-data retraining [13]. Our current-checkpoint and historical-snapshot results must consequently be read as different threat settings.

Prior work also limits the novelty available to a small pilot. Yuan et al. evaluate continual unlearning and additional output dimensions [14], while Hu et al. examine the reversibility of apparent forgetting under benign relearning [6]. Repeated requests and a difference between output behavior and retained information are not new discoveries. The contribution here is a transparent diagnostic sequence: establish subject-specific learning, match the answer inventory of untrained controls, make two actual retained-only updates, and preserve all resulting observations. The study provides a compact artifact for developing more economical audits and identifying invalid positive controls before a larger evaluation.

The initial motivation is practical. A one-year study should not spend its audit budget on models that never learned the target association, or treat familiar answer tokens as proof of subject-specific inclusion. We therefore ask: after original associations have been demonstrated, does cheap continuation on retained records remove exact greedy

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extraction and known-candidate association evidence together? We also ask whether a favorable result survives a fresh retained-data reference and an explicit change in attacker checkpoint access. Our answer is descriptive and narrow: the tested continuation baseline often suppresses extraction while leaving stronger association signals.

The author's earlier SemEval work explored fine-tuning and distillation for selective unlearning [1]. That method is background and a future comparator; it was not implemented in this pilot. Likewise, LoRA [5] motivates economical adaptation, but the successful association control below uses partial-dense adaptation. The failed LoRA runs remain part of the evidence.

2 Methods

2.1 Synthetic association unit and controls

Each corpus contains 56 fictional subjects. Groups A, B and R contain eight trained subjects each; A and then B are designated for deletion, and R remains throughout. Four additional groups, cal_A, cal_B, test_A and test_B, each contain eight untrained subjects. Each A code is reused once in its calibration group and once in its test group; the same construction applies to B. Subjects differ while the code inventories match exactly. The membership unit is a subject–code pair, not exposure to a digit or to a code's marginal distribution.

Codes are sampled without replacement as six-digit strings, including leading zeros. Subject identifiers combine a corpus seed and randomly selected word pairs. Two training question wordings request each subject's code. Evaluation uses the first wording and a third, held-out wording. All views of a subject remain together. The corpus and training-order seeds are 211 and 307; the pretrained checkpoint is fixed, so these are not independent pretrained initializations. This small design deliberately favors traceability over natural-language realism.

2.2 Development gate and adaptation recipe

The primary checkpoint is the official Qwen3-0.6B release [10], snapshot `c1899de289a0` (full revision in the artifact). MLX-LM 0.32.2 and MLX 0.32.4 provide training and inference [9]. Native chat completions are prompt-masked; the actual tokenizer boundaries are recorded. Greedy generation uses a non-thinking assistant prefix, temperature zero and at most 12 generated tokens. A result counts as extraction only when whitespace stripping yields exactly the correct six digits. No digit repair or answer substitution is used. Training caps sequences at 192 tokens, uses one batch per optimizer update (gradient accumulation one), and enables activation checkpointing. The tested LoRA configurations use scale 20 and dropout zero.

Development seed 101 tests the positive learning control before any locked sequence. Passing requires at least 21/24 correct unique top-1 associations under the exact wording, at least 12/24 under the held-out wording, and at least 6/8 exact associations in each trained group. Candidate ties within 10^{-6} fail. The initially specified A/B runs failed. A documented development-only amendment then specified C/D in a fixed order; no locked corpus was inspected to select them. Table 1 retains every attempt. Recipe D was selected before the locked runs and updates dense weights in the last four decoder blocks, with other weights frozen. It is neither a LoRA adapter nor full-model fine-tuning. Differences in rank, learning rate, exposure and batch size prevent a causal claim that dense adaptation is intrinsically better.

2.3 Two actual updates and fresh references

For each locked corpus, an original model is adapted on $A \cup B \cup R$ using D. The original association gate must pass before a deletion comparison is interpreted. Stage one resumes these adapted weights and trains only on $B \cup R$ for 64

Table 1. Development association checks. Exact and paraphrase columns count correct subject-specific candidate rankings out of 24. D passed the predeclared gate; failed recipes do not establish deletion efficacy. Exposures count examples per subject across two training wordings.

Recipe	Mechanism	Batch	Learning rate	Exposures	Exact	Para.
A	LoRA rank 16	1	5×10^{-4}	24	3	3
B	LoRA rank 32	1	10^{-3}	48	3	3
C	LoRA rank 32	4	10^{-4}	128	8	3
D	Last four blocks, dense	4	5×10^{-5}	256	24	22

Adam updates at batch four. Stage two resumes the stage-one weights and trains only on R for another 64 updates. Both continuations retain D's learning rate 5×10^{-5}. The optimizer is freshly initialized at each stage; old optimizer state is not resumed. This is inexpensive retained-only continuation, used as a baseline rather than presented as an unlearning method.

At each stage, a separate reference starts from the same pretrained checkpoint and trains on the remaining records with D's 256 exposures per subject. The original, first reference and second reference thus receive 1536, 1024 and 512 updates, respectively. This equal-per-record recipe is a reference for the defined adaptation pipeline, not retraining of the original model's pretraining corpus. A comparison passes the narrow retained-utility gate only if every still-retained group has at least 7/8 exact candidate top-1 associations. Exact generation, paraphrase ranking and paraphrase generation remain separately recorded. They are not absorbed into that gate.

A no-deletion checkpoint is retained. A second control actually applies a known-ID output postprocessor to its saved generations: designated deleted IDs receive `[DELETED]`. This produces zero deleted-code extraction without changing model weights. It is an endpoint-control demonstration, not a deployed access-control service or a security defense for released weights.

2.4 Fixed-recipe replay and interruption lineage

We additionally transfer D unchanged to the official Qwen3.5-0.8B text component [11], snapshot `2fc06364715b` (full revision in the artifact). Its native MLX text wrapper excludes the vision component; the last four text decoder blocks, layers 20–23, supply 53 trainable tensors. One development association sanity check is allowed, followed by the same two corpus/order seeds if it passes. These corpora were already examined with the primary checkpoint. Consequently, this is an exploratory cross-model replication, not a new confirmatory sample or a controlled comparison of release age. Model family, scale, architecture, post-training, trainable parameter count and native formatting change together.

An external tool restart interrupted the second model during seed 307's first fresh-reference training. A separately recorded recovery protocol reused hash-checked completed seed-211 states and the seed-307 original, then restarted only the unfinished reference from the same pretrained checkpoint with the same fixed recipe before completing the remaining stages. No new recipe or outcome-based selection was introduced. The original partial run remains preserved separately; the artifact identifies the canonical recovered result and its source hashes.

2.5 Scores, calibration and access

For candidate association (s, c), let $\ell_\theta(s, c)$ be the mean answer-token negative log likelihood. Each subject has a known pool C of eight candidate codes. The association margin is

$$m_\theta(s, c) = \frac{1}{7} \sum_{c' \in C \setminus \{c\}} \ell_\theta(s, c') - \ell_\theta(s, c). \tag{1}$$

Candidate top-1 requires the correct code to beat all seven alternatives. This is a known-candidate discrimination test, not black-box reconstruction of an unknown secret. Membership scoring assumes that an auditor can test a proposed subject–code pair and candidate inventory. Greedy extraction requires no candidate list.

All attack scores first average the exact and paraphrase views within each subject. We retain current negative likelihood and current association margin; gains additionally require the public pretrained checkpoint. Snapshot-delta scores require the private original checkpoint and measure $\ell_{\theta_t} - \ell_{\theta_0}$. This extra historical access is explicitly separate from current-checkpoint access. Reported association AUROC compares deleted subjects with their matched test controls, assigning half credit to ties. The maximum calibration-control score is a fixed threshold; detection and false-positive counts for every attack remain in the raw artifact. Stage one has eight positive, eight calibration and eight test subjects; stage two has 16 of each. These counts cannot establish a low-false-positive operating point or certify privacy.

The implementation preserves frozen protocol and source hashes, training configurations, synthetic oracles, raw candidate losses, raw generations and serving-control outputs. Saurav K. Aryal directed the experiments with Codex-assisted implementation and execution in the shared local workspace. Separate metadata annotations correct an inherited "fresh LoRA" label in generic training records; saved tensor keys and configuration identify D's actual partial-dense mechanism. The immutable records are retained rather than silently rewritten.

3 Results

Both primary locked originals passed with 24/24 exact and 23/24 held-out-paraphrase candidate associations. All original, fresh-reference and warm-continuation arms preserved 8/8 exact associations in each remaining group at both deletion stages. This establishes the intended narrow comparison, while leaving broader language utility untested.

Extraction and association diverge. After deleting A and then B, exact greedy extraction is 0/16 for both Qwen3-0.6B warm models. Nevertheless, each ranks the correct deleted code above the seven alternatives for 11/16 subjects. Fresh R-only references rank 3/16 and 2/16 correctly. Current association-margin AUROC is 0.8164 and 0.8477 for warm continuation, versus 0.4844 and 0.5625 for fresh references (Figure 1). These observations demonstrate a disagreement between specified tests, not a recovered unknown-secret attack or universal residual-memory claim. At stage one, warm exact extraction is 1/8 and 0/8, while candidate top-1 remains 7/8 in both corpora; current association AUROC is 0.8906 and 0.7969.

Prior access changes the question. Fresh Qwen3-0.6B references, which never adapt on A or B, still have final snapshot-delta AUROC 0.8203 and 0.8516 when the original adapted snapshot is available. Their current-checkpoint association AUROC is near chance by comparison. This does not show a post-only failure of fresh retraining: the historical checkpoint was explicitly granted to that attacker. Final warm snapshot-delta AUROC is 0.5430 and 0.6133, so no single attack dominates all access settings. Table 2 retains these distinct outcomes rather than reporting the best score alone.

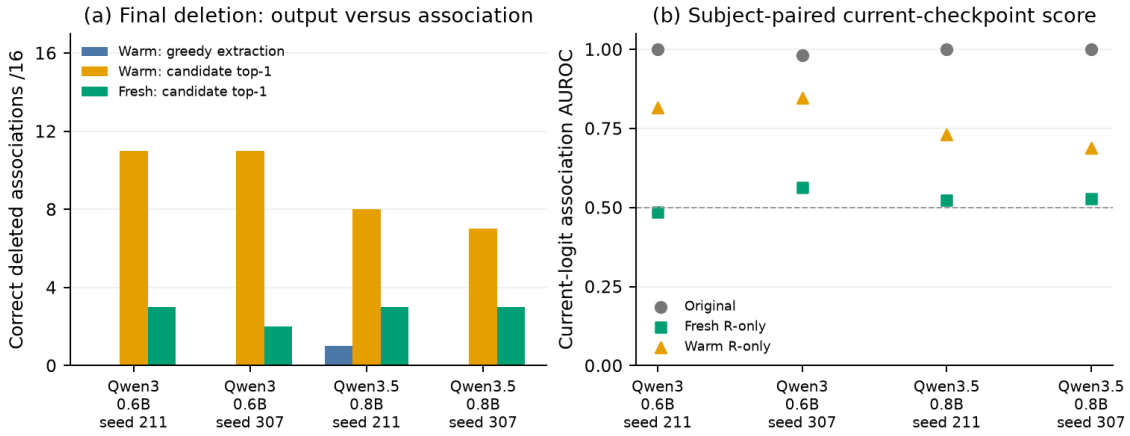


Fig. 1. Final deletion diagnostics generated from completed raw results. Left: warm retained-only continuation can suppress exact greedy extraction while retaining known-candidate associations. Right: current-checkpoint association discrimination against matched untrained subjects. Each point or bar summarizes one fixed corpus/order-seed condition; the paired subjects, views and repeated arms do not supply independent model runs.

Table 2. Final-stage counts and historical-snapshot scores. Extraction and candidate top-1 concern 16 deleted subjects. "R para." is greedy extraction on eight retained subjects under the held-out wording. Every displayed arm retains 8/8 exact R candidate associations. Snapshot AUROC grants access to both original and current models.

Model / seed	Final arm	Extraction	Top-1	R para.	Snapshot AUROC
3-0.6B / 211	Original	16/16	16/16	7/8	0.500
3-0.6B / 211	Fresh R	0/16	3/16	8/8	0.820
3-0.6B / 211	Warm R	0/16	11/16	6/8	0.543
3-0.6B / 307	Original	16/16	16/16	5/8	0.500
3-0.6B / 307	Fresh R	0/16	2/16	7/8	0.852
3-0.6B / 307	Warm R	0/16	11/16	5/8	0.613
3.5-0.8B / 211	Original	16/16	16/16	6/8	0.500
3.5-0.8B / 211	Fresh R	0/16	3/16	8/8	0.723
3.5-0.8B / 211	Warm R	1/16	8/16	6/8	0.453
3.5-0.8B / 307	Original	16/16	16/16	6/8	0.500
3.5-0.8B / 307	Fresh R	0/16	3/16	8/8	0.742
3.5-0.8B / 307	Warm R	0/16	7/16	7/8	0.465

The utility qualification matters. Final retained paraphrase extraction for the primary warm models is 6/8 and 5/8; their fresh references achieve 8/8 and 7/8. Even the original models differ at 7/8 and 5/8. Thus, the experiment matches exact association utility while failing to establish matched paraphrase generation or broader task quality (Figure 2). It would be misleading to describe these results as forgetting at unchanged general utility.

Fixed-recipe replay on a second model. The Qwen3.5-0.8B development sanity check passed 24/24 exact and 24/24 paraphrase associations before the fixed-corpus sequences. Its originals achieved 24/24 exact associations and 23/24 and 24/24 held-out-wording associations, respectively. Seed 211 ended with 1/16 greedy extractions and 8/16 candidate

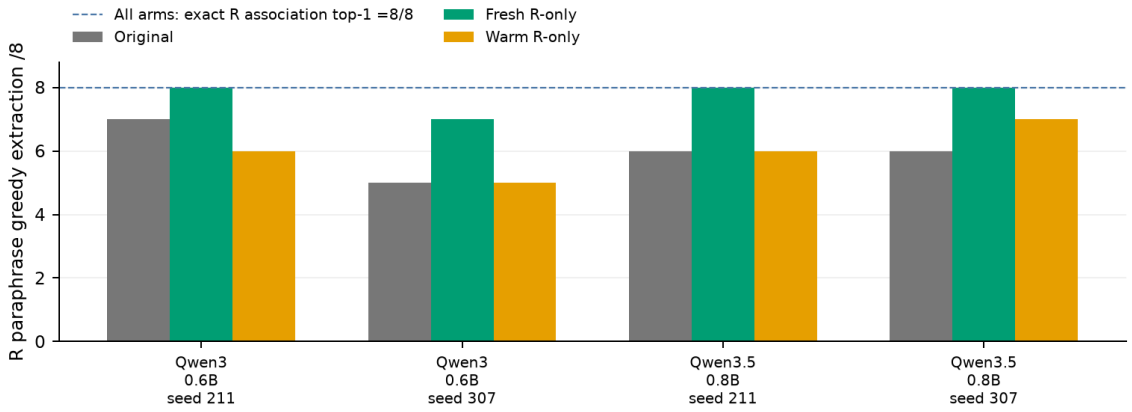


Fig. 2. Final retained utility has multiple meanings. All arms reach the exact association ceiling, but held-out-wording greedy extraction varies. This small paired diagnostic is a utility warning, not an estimate of general language quality.

top-1 associations; current-binding AUROC was 0.7305, versus 0.5234 for its fresh reference. Seed 307 ended with 0/16 greedy extractions and 7/16 candidate top-1 associations; current-binding AUROC was 0.6875, versus 0.5273 for its fresh reference. Both warm models and fresh references retained 8/8 exact R associations. Warm R paraphrase generation was 6/8 and 7/8, versus 8/8 for both fresh references, so broader utility remains unmatched. All displayed final arms passed the exact R utility gate. This is exploratory replication on the same synthetic corpora, with family, parameter scale, architecture, post-training, trainable parameter count and native supervised-formatting offsets changing together. It does not isolate an effect of model age. The newer-model training wrapper has two formatting tokens before the answer, compared with four for Qwen3; native masked training losses are therefore not comparable units.

Scope of the evidence. Two synthetic corpora, eight controls per deletion group, short digit strings and a fixed pretrained initialization sharply limit generalization. Corpus and optimizer-order changes are coupled. The closed-set task supplies stronger candidate knowledge than an unconstrained extraction setting. The continuation baseline is intentionally cheap; the study neither compares state-of-the-art unlearning methods nor establishes the mechanism of residual information. No adaptive query allocator, low-FPR claim, attack-success confidence interval or privacy certificate is supported. Model timing on a shared host and MLX scoring allocator peaks exclude full training-memory accounting and cannot establish target-device efficiency.

4 Future Work

The first next step is positive-control readiness for economical adaptation. LoRA should establish reliable subject-specific learning on development data before it is used for deletion claims. A larger study should compare D with a reproducibly binding LoRA configuration, the author's fine-tuning/distillation method, and methods already available in OpenUnlearning [1, 2]. Public TOFU and MUSE compatibility tests should remain separate from newly generated controlled records; neither public text provenance nor this digit-string task represents the full privacy setting.

The second step is a bounded sequential audit with explicit cost and access. Longer request sequences should preserve discovery, calibration and untouched confirmation subjects, hold out generation templates, and evaluate at matched retained-task quality as well as matched deletion compute. A frozen extensive attack battery can identify

failure conditions, after which budgeted discovery policies are compared by the cost to recover those same failures. Costs must include shadow training, attack discovery, calibration and inference, with first-use and amortized accounting. Prior/current-snapshot attacks require their own access label throughout.

Cost reduction alone is not a new auditing principle. Guo et al. already combine cheap screening and selective expensive verification in an incentive model [4]; their hierarchy is a required allocation comparator, not a novelty claim for this project. Kutta et al. provide sequential sample-efficient f-DP auditing with statistical guarantees [7]. Our proposed empirical deletion-failure study does not inherit those guarantees. Larger negative-control populations, repeated training runs and valid treatment of dependent stages are prerequisites for calibrated operating-point claims. A useful result may remain "inconclusive" when original binding, utility matching or the audit budget is insufficient.

5 Conclusion

This pilot demonstrates an auditable discrepancy: zero exact greedy extraction can coexist with correct known-candidate association rankings and current-checkpoint inclusion discrimination after two retained-only updates. The conclusion rests on verified original association learning, matched answer inventories and fresh retained-data references. It is bounded by a partial-dense adaptation recipe, synthetic short strings, imperfect paraphrase utility and small dependent samples. The artifact supports a larger access-aware sequential evaluation; it does not establish LoRA unlearning, a novel deletion method or privacy.

Artifact availability. The artifact repository is https://github.com/Saurav-K-Aryal/sequential-deletion-audit-pilot. It includes figure generation, source hashes and a CPU-only verifier. Cached pretrained weights and adaptation weights are excluded from the release; local retraining requires a separately obtained official checkpoint.

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